

# A Unified Theory of Electronic Logical-Structural and Physical Properties Based on the Minimal Instantiable Topological Scale

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## Abstract

Within the full SRE-Dynamics framework, presuppositions of material entities are discarded; space, time and energy are not taken as pre-given background. This paper introduces the **minimal instantiable topological scale**  $l_{min}$ , corresponding to the emergent ontological ultraviolet boundary (Planck length) of the physical rendering layer. It is proposed that the electron is not a pre-existing material entity in the underlying causal space. Instead, it is an irreducible self-consistent topological closed loop generated within the physical rendering layer via homomorphic mapping, after causal interactions undergo dissipation-compensation accounting and screening by multi-scale rigid boundaries.

By back-propagating the topological frequency and book-keeping ratios of this topological closed loop under reciprocal-measurement sampling, it can be demonstrated that rest mass, unit electric charge, spin-1/2, and the uncertainty principle are all inevitable observable effects produced by this topological closed loop within the physical rendering layer.

## I. Core Definition: Minimal Instantiable Topological Scale of the Rendering Layer

In the SRE framework, physical reality is the instantiated rendering result of causal space screened by multi-scale rigid boundaries and processed via homomorphic mapping.

Note:  $l_{min}$  is not an intrinsic native discrete step of the underlying causal network. The underlying causal space permits smaller logical differences; interactions falling below the instantiation boundary remain in the pending-for-instantiation state and cannot be projected into the physical rendering layer.

- Minimal instantiable topological scale  $l_{min}$ :** The lower scale bound for topological events to complete instantiation inside the physical rendering layer. It corresponds to the **emergent ontological ultraviolet boundary (Planck length)** in SRE axioms and sets the resolution lower limit for reciprocal-measurement book-keeping within the rendering layer.
- Derivative Spacetime:** Spatial metrics and temporal increments are both topological book-keeping products of dissipation-compensation duality driven by reciprocal measurement.  $l_{min}$  defines the minimum effective scale output by this book-keeping system.

## II. Logical-Topological Formulation of the Electron: N-Scale Self-Consistent Topological Closed Loop

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No pre-packaged electronic “algorithm unit” exists in the underlying causal space. The electron is a topological structure emerging within the physical rendering layer after iterative dissipation-compensation of numerous asynchronous causal interactions, once instantiation conditions are satisfied. It consists of closed-link segments built from  $N$  minimal instantiable topological scales  $l_{min}$ .

- **Topological Closure:** The rendering-layer causal sequence  $[A_1, A_2, \dots, A_N]$  possesses closed-loop back-coupling characteristics:  $A_N$  establishes causal association back to  $A_1$ , forming a self-consistent topological closed loop. Internal reciprocal-measurement and dissipation-compensation accounting persist inside the loop, maintaining overall self-consistency of residuals.
- **Möbius-phase Characteristic:** This closed loop exhibits a two-layered phase topology. To return topological residuals to their initial symmetric state,  $2N$  reciprocal-measurement triggers at the  $l_{min}$ -scale must be completed; phase restoration cannot be achieved after only  $N$  cycles.

## III. Parametric Verification of Observable Physical Properties

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Taking the rendering-layer minimal instantiable topological scale  $l_{min}$  as the scale benchmark, each observable physical property of the electron may be interpreted as emergent effects of this topological closed loop under reciprocal-measurement sampling.

### 1. Rest Mass: Measure of Topological Processing Overhead of the Closed Loop

- **Logical-Topological Definition:** Rest mass  $m$  characterises the additional topological book-keeping overhead that reciprocal-measurement protocols must expend to process this topological closed loop.
- **Physical Interpretation:** When reciprocal measurement attempts to gauge or displace this electronic closed loop within the rendering layer, full pre-processing of dissipation-compensation for the  $N$  internal topological segments must first be completed. This  $l_{min}$ -based topological processing load manifests physically as inertial effects.
- **Corollary:**  $m_e \propto N \cdot l_{min}$ . The experimentally observed constant mass of the electron originates from structural stability of the closed-loop depth  $N$  of this emergent topological closed loop.

### 2. Unit Electric Charge: Characteristic Signature of Topological-Step Bias of the Closed Loop

- **Logical-Topological Definition:** Electric charge  $e$  is the constant topological logical bias exerted by this electronic topological closed loop onto external causal links during each round of  $l_{min}$ -scale reciprocal-measurement cycle.
- **Physical Interpretation:** Coupled with surrounding rendering-layer topological structures via reciprocal-measurement protocols mediated by Möbius light-residual manifolds, the electronic closed loop continuously introduces fixed offsets to neighbouring causal associations and generates logical potential differences at the topological level.
- **Corollary:** Unit electric charge represents a persistent topological modification effect at the  $l_{min}$  instantiable scale, macroscopically manifesting as electromagnetic interaction.

### 3. Spin-1/2: Phase-Locking of the Möbius Closed Loop

- **Logical-Topological Definition:** Spin arises from the reciprocal-measurement book-keeping beat required for the topological closed loop to complete global causal-consensus settlement.
- **Physical Interpretation:** Constrained by the Möbius two-layer topology,  $2N$  reciprocal-measurement trigger cycles are required for topological residuals of the closed loop to return to the initial symmetric configuration.
- **Corollary:** This topological behaviour, which demands a doubled cycle count for phase restoration, yields fermionic spin-1/2 observational behaviour at the statistical level.

### 4. Uncertainty Principle: Incomplete-Sampling Effect from Instantiation Boundaries

- **Logical-Topological Definition:** The uncertainty relation originates from the fact that reciprocal-measurement sampling itself consumes  $l_{min}$ -scale instantiation resources; complete synchronous sampling cannot be performed on a finite-depth topological closed loop.
- **Physical Interpretation:** Any single reciprocal-measurement sampling consumes at least one unit of  $l_{min}$ -scale book-keeping resource. The internal topological depth  $N$  of the electronic closed loop greatly exceeds the instantiation resolution of a single sampling. When sampling anchors the position (topological index), the cyclic frequency (corresponding to momentum) cannot be fully resolved simultaneously.
- **Corollary:** The uncertainty principle is an incomplete-sampling effect of reciprocal-measurement imposed by the emergent ontological ultraviolet boundary.

## IV. Conclusion: Particles as Topological-Rendered Products after Homomorphic Mapping

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From the above deductions we arrive at the core proposition: **All intrinsic constants experimentally measured for the electron are statistical emergent manifestations after global reciprocal-measurement protocol sampling of a self-consistent topological closed loop constructed from  $N$ -fold minimal instantiable topological scale  $l_{min}$ .**

1. **The electron possesses no underlying material substance:** It is not a primitive object of causal space. Instead, it is a self-consistent topological closed-loop structure within the rendering-layer that resists simple linear expansion, after screening by multi-scale rigid boundaries and homomorphic mapping.
2. **Particle stability originates from self-consistency of the topological closed loop:** As long as dissipation-compensation accounting inside the closed loop maintains global self-consistency and residual conditions remain satisfied, this topological object acquires logical stability within the physical rendering layer.
3. **Unification:** All intrinsic physical properties of the electron can be reduced to topological reciprocal-measurement book-keeping relations formulated in terms of  $l_{min}$ .

## V. Closing Remarks

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This paper demonstrates that within the SRE-Dynamics framework, a unified topological-logical account for all known observational behaviours of the electron can be achieved via the minimal instantiable topological scale  $l_{min}$ . This suggests that selected postulates of particle physics may be transformed into deductive inferences of causal-topology, dissipation-compensation and reciprocal measurement.

Full numerical simulation of this topological structure and benchmarking against the hierarchical dissipative binary-network model will be presented in subsequent works of this series.